Book Notes

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- 3 MODELS
  - VERY LONG RUN:
    - which focuses on the growth of the economy's capacity to produce goods and services
    - historical accumulation of capital and improvements in technology
  - LONG RUN
    - <u>Fixed capital and technology</u> determine the productive capacity of the economy—we call this capacity "potential out put."
    - Potential Output= supply of goods + services
    - Prices and inflation over this horizon are determined by fluctuations in demand
  - SHORT RUN:
    - fluctuations in demand determine how much of the available capacity is used and thus the level of output and unemployment
    - In contrast to the long run, in the short run prices are relatively fixed and output is variable.

VERY LONG RUN

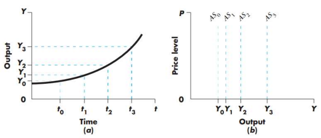
- Growth theory
- accumulation of inputs—investment in machinery, for example— and improvements in technology lead to an increased standard of living
- IGNORE recessions and booms and short run fluctuations in employement
- These short term fluctuations tend to average out over the years
- 

FIGURE 1-3 DETERMINATION OF AGGREGATE SUPPLY: THE VERY LONG RUN.

- In industrialized countries, standard of living increases by devlopment of new tech and accumulation of capital
- In developing, development of well functioning infrastructure is more important than the devlopment of new tech

LONG RUN

- Level of output determined by productive capacity of the economy
- Price level determined by level of demand relative to output economy can supply
- Aggregate supply= amount of output the economy can produe given the resources and tech available
- Aggregate demand= goods to consume for new investment+ goods purchased by govt + good exported outside
- 

FIGURE 1-2 AGGREGATE DEMAND AND SUPPLY: THE LONG RUN.

- Depicts for each given price level, the quanity of output firms are willing to supply.
- Depends upon productive capacity of the economy
- price level in the economy and total output. **The *aggregate supply (AS) curve* depicts, for each given price level, the quantity of output firms are willing to supply**. The position of the aggregate supply curve depends on the productive capacity of the economy. **The *aggregate demand (AD) curve* presents, for each given price level, the level of output at which the goods markets and money markets are simultaneously in equilibrium.** The position of the aggregate demand curve depends on monetary and fiscal policy and the level of consumer confidence. The intersection of aggregate supply and aggregate demand determines price and quantity.[1]
- **In the long run, the aggregate supply curve is vertical.** (Economists argue over whether the long run is a period of a few quarters or of a decade.) Output is pegged to the position where this supply curve hits the horizontal axis. The price level, in contrast, can take on any value.
- Mentally shift the aggregate demand schedule to the left or right. You will see that the intersection of the two curves moves up and down (the price changes), rather than horizontally (output doesn't change). **It follows that *in the long run* output is determined by aggregate supply alone and prices are determined by both aggregate supply and aggregate demand.** This is our first substantive finding.
- The growth theory and long-run aggregate supply models are intimately linked: The position of the vertical aggregate supply curve in a given year equals the level of output for that year from the very long-run model, as shown in Figure 1-3. Since economic growth over the very long run averages a few percent a year, we know that the aggregate supply curve typically moves to the right by a few percent a year.[2]
- We are ready for our second conclusion: **Very high inflation rates—that is, episodes with rapid increases in the overall price level—are always due to changes in aggregate demand**. The reason is simple. Aggregate supply movements are on the order of a few percent; aggregate demand movements can be either small or large. So the only possible source of high inflation is large movements of aggregate demand sweeping across the vertical aggregate supply curve. In fact, as we will eventually learn, the only source of really high inflation rates is government-sanctioned increases in the money supply.[3]

SHORT RUN

Examine panel *(b)* in Figure 1-1. When we take a magnified look at the path of output, we see that it is not at all smooth. Short-run output fluctuations are large enough to matter a great deal. Accounting for short-run fluctuations in output is the domain of aggregate demand.[4]

The mechanical aggregate supply–aggregate demand distinction between the long run and the short run is straightforward. *In the short run, the aggregate supply curve is flat.* The short-run aggregate supply curve pegs the price level at the point where the supply curve hits the vertical axis. Output, in contrast, can take on any value. The underlying assumption is that the level of output does not affect prices in the short run. Figure 1-4 shows a horizontal short-run aggregate supply curve.

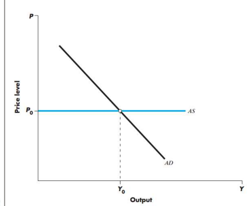


FIGURE 1-4 AGGREGATE DEMAND AND SUPPLY: THE SHORT RUN.

Repeat the exercise above and mentally shift the aggregate demand schedule to the left or right. You will see that the intersection of the two curves moves horizontally (output changes), rather than up and down (the price level doesn't change). **It follows that *in the short run* output is determined by aggregate demand alone and prices are unaffected by the level of output.** This is our third substantive finding.[5]

Much of this text is about aggregate demand alone. We study aggregate demand because in the short run aggregate demand determines output and therefore unemployment. When we study aggregate demand in isolation, we are not really ignoring aggregate supply; rather, we are assuming that the aggregate supply curve is horizontal, implying that the price level can be taken as given.

MEDIUM RUN

answer is that when high aggregate demand pushes output above the level sustainable according to the very long run model, firms start to raise prices and the aggregate supply curve begins to move upward. The *medium run* looks something like the situation shown in Figure 1-5; the aggregate supply curve has a slope intermediate between horizontal and vertical. **The question, How steep is the aggregate supply curve?, is in effect the main controversy in macroeconomics.**

It turns out that prices usually adjust pretty slowly; thus, over a 1-year horizon, changes in aggregate demand give a good, though certainly not perfect, account of the behavior of the economy. **The speed of price adjustment is summarized in the *Phillips curve*, which relates inflation and unemployment, one version of which is shown in Figure 1-6.**

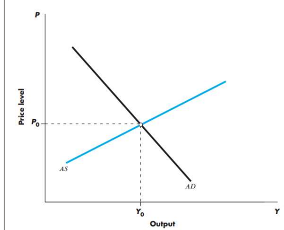


FIGURE 1-5 AGGREGATE DEMAND AND SUPPLY.

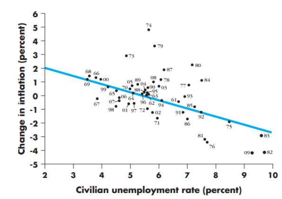


FIGURE 1-6 UNEMPLOYMENT AND THE CHANGE IN INFLATION, 1961–2009.
(Source: Bureau of Labor Statistics and International Financial Statistics, IMF.)

GROWTH AND GDP

- Growth rate of economy is the rate at which the gross domestic product is increasing

What leads to increase in GDP?

① change in resources → Growth in GDP

Increased labour force (labour) → increased production → increase in GDP.
Increased buildings & machine (capital) → increased production

② Efficiency of factors of production may change

i) more knowledge
ii) more tech

Business Cycle

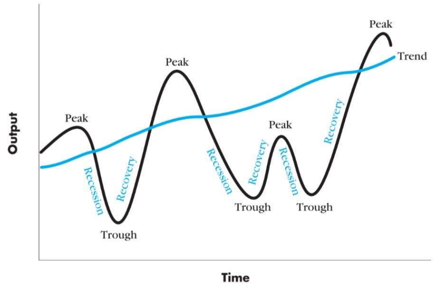
- Regular pattern of expansion (recovery) and contraction (recession) in economic activity around path of trend growth.

- Cyclical peak → high economic activity
- Cyclical trough → low " "

GDP relative to trend over the business cycle.

The gold line in Figure 1-7 shows the *trend path of real GDP*. **The trend path of GDP is the path GDP would take if factors of production were fully employed.** Over time, GDP changes for the two reasons we already noted. First, more resources become available: The size of the population increases, firms acquire machinery or build plants, land is improved for cultivation, the stock of knowledge increases as new goods and new methods of production are invented and introduced. This increased availability of resources allows the economy to produce more goods and services, resulting in a rising trend level of output.

But, second, factors are not fully employed all the time. Full employment of factors of production is an economic, not a physical, concept. Physically, labor is fully employed if everyone is working 16 hours per day all year. In economic terms, there is full employment of labor when everyone who wants a job can find one within a reasonable amount of time. Because the economic definition is not precise, we typically define full employment of labor by some convention, for example, that labor is fully employed when the unemployment rate is 5 percent. Capital similarly is never fully employed in a



Recession → Period between peak & trough

Recovery → Period between trough & peak

Deviations from output → Output Gap.
↓
Gap between actual output & output the economy could produce at full employement given existing resources. (potential output).

Output gap = Actual output − potential output

Inflation

- Expansionary aggregate demand policies tend to produce inflation, unless they occur when the economy is high levels of unemployement

Inflation measure: **rate of** change of consumer price index (CPI)

Inflation, like unemployment, is a major macroeconomic concern. However, the costs of inflation are much less obvious than those of unemployment. In the case of unemployment, potential output is going to waste, and it is therefore clear why the reduction of unemployment is desirable. In the case of inflation, there is no obvious loss of output. It is argued that inflation upsets familiar price relationships and reduces the efficiency of the price system. Whatever the reasons, policymakers have been willing to

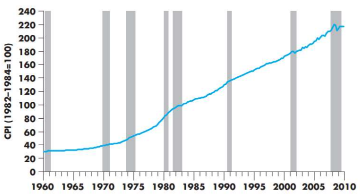


FIGURE 1-10 CONSUMER PRICE INDEX, 1960–2009.
(Source: Bureau of Labor Statistics.)

Growth → Accumulation of economic inputs

Aggregate supply → Growth (long term)

Aggregate demand → Monetary Policy, interest rates & expectations & by fiscal policy

Chapter- 2

Output

Production (Output)
- Factor payments
  - $\boxed{\text{Wages to labour}}$ 75%
  - $\boxed{\text{capital (interest \& dividends)}}$ 20%
- Growth & aggregate supply

Demand (Input)
- Consumption
- Invested for future
- Aggregate demand

GDP → Value of final goods & services produced in country within a given period.

Production function

GDP = f(labour, capital)
Y N K

$$\boxed{Y = f(N, K)}$$

$$\boxed{GDP = \text{Labour Payments} + \text{Capital Payments} + \text{Profit}}$$

$$\boxed{Y = (w \times N) + (r \times K) + \text{Profit}}$$

w → wage

r → rate of capital payment

GNP
Gross National Product

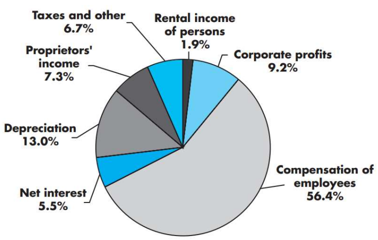
Factor payments include receipts from abroad made as factor payments to domestically owned factors of production

Net Domestic Product

- Capital wears out, depreciates while being used to produce output.

$$\boxed{NDP = GDP - \text{depreciation}}$$

Total value of production − Amount of capital used in production (About 11%).



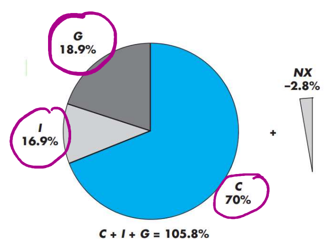
(a) Payments to factors of production

NATIONAL INCOME

The third complication is that businesses pay indirect taxes (i.e., taxes on sales, property, and production) that must be subtracted from NDP before making factor payments. These payments are large, amounting to nearly 10 percent of NDP, so we need to mention them here. (Having done so, we won't mention them again.) What's left for making factor payments is *national income*, equaling about 80 percent of GDP.

Aggregate Demand

- Consumption spending by households C
- Investment spending by businesses & households I
- government (federal, state, local) purchases of goods and services G
- foreign demand for our net exports NX



(b) Components of demand for output

FIGURE 2-1 COMPOSITION OF U.S. GDP IN 2009.
(Source: Bureau of Economic Analysis.)

$$\boxed{Y = C + I + G + NX}$$

Y → demand for domestic input

→ fundamental National Income Accounting Identity.

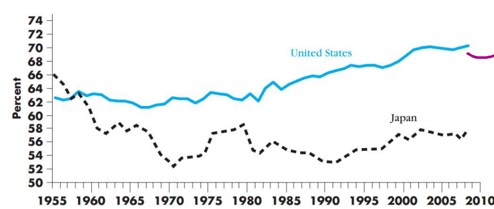


FIGURE 2-2 CONSUMPTION AS A SHARE OF GDP: UNITED STATES AND JAPAN, 1955–2008.
(Source: International Financial Statistics, IMF.)

higher consumption
↓
less saving
↓
less investment
larger trade deficits

Aggregate Demand

- Consumption spending by households
  C
- Investment spending by businesses & households
  I
- government (federal, state, local) purchases of goods and services
  G
- foreign demand for our net exports
  NX

eign goods and foreign spending on domestic goods. When foreigners purchase goods we produce, their spending adds to the demand for domestically produced goods. Correspondingly, that part of our spending that purchases foreign goods has to be subtracted from the demand for domestically produced goods. Accordingly, the difference between exports and imports, called *net exports*, is a component of the total demand for our goods. U.S. net exports have been negative since the 1980s, as shown in Figure 2-4,

imports added in consumption column & subtracted from net imports.

- housing construction
- building of machinery
- construction of offices & factories

- Any activity that increases economic ability to produce output in future + Human capital
  ↓
  knowledge & ability

- Adding physical stock of capital including inventories in business sectors.

$$\text{Net Investment} = \text{Gross Investment} - \text{Depreciation}$$

Government Spending:

1. Called as purchases of goods & services
2. Transfer Payments: Payments made to people without their providing current service in exchange.
   - Could be social security benefits and unemployment benefits.
   - NOT counted as part of GDP because transfers are not part of current production.
   - Plays a large role in determining how economy is split between public and private sector.

SIMPLE ECONOMY

1. No government + foreign trade
2. Output → Y
   Consumption → C
   Investment → I

$\boxed{\text{Output produced} = \text{Output Sold}}$

③ Output unsold → inventory → accumulation of inventories $= \boxed{\text{investment}}$

$\boxed{Y \equiv C + I}$ → Investment

$\boxed{Y \equiv S + C}$ → consumption

private sector saving

$\boxed{C + I \equiv Y \equiv S + C}$

demand

allocation of income.

$\boxed{I = Y - C \equiv S}$

With Government & foreign Trade

① Government purchases of goods & services → G

Taxes → TA

Transfers to private sector (including interest on public debt) → TR

Net Exports → NX

$Y = C + I + G + NX$

But a) Part of income spent on taxes

b) Private sector receives net transfers

$\boxed{\text{Disposable income} = Y + TR - TA} = S + C$

$\Rightarrow YD - TR + TA \equiv C + I + G + NX$

$\Rightarrow \cancel{C} + S - TR + TA \equiv \cancel{C} + I + G + NX$

$\Rightarrow S - I \equiv G + NX + TR - TA.$

$\Rightarrow \boxed{S - I = (G + TR - TA) + NX}$

taxes govt receives

Net Exports / Trade Surplus

total govt

government budget deficit

Exports / Trade Surplus

total govt expenditure (BD)

government budget deficit

Negative Budget surplus.

How to dispose savings?

① loans to govt

② lend to foreigners

③ lend to business firms. (funds for investment)

So as to not DOUBLE COUNT in GDP, value of intermediate goods is not added, but only the final good value

Why GDP is not true measure of growth?

- Some outputs are poorly measured because they are not traded in the market. If you bake homemade pie, the value of your labor isn't counted in official GDP statistics. If you buy a (no doubt inferior) pie, the baker's labor is counted. This means that the vastly increased participation of women in the labor force has increased official GDP numbers with no offsetting reduction for decreased production at home. (We officially measure the value of commercial day care, but taking care of your own kids is valued at zero.)

  Note, too, that government services aren't directly priced by the market. The official statistics assume that a dollar spent by the government is worth a dollar of value.[8] GDP is mismeasured to the extent that a dollar spent by the government produces output valued by the public at more or less than a dollar.
- Some activities measured as adding to GDP in fact represent the use of resources to avoid or contain "bads" such as crime or risks to national security. Similarly, the accounts do not subtract anything for environmental pollution and degradation. This issue is particularly important in developing countries. For instance, one study of Indonesia claims that properly accounting for environmental degradation would reduce the measured growth rate of the economy by 3 percent.[9]

- It is difficult to account correctly for improvements in the quality of goods. This has been the case particularly with computers, whose quality has improved dramatically while their price has fallen sharply. But it applies to almost all goods, such as cars, whose quality changes over time. The national income accountants attempt to adjust for improvements in quality, but the task is not easy, especially when new products and new models are being invented.

REAL GDP

Real GDP measures changes in physical output in economy b/w different time periods by valuing all goods produced in 2 periods at same prices or in constant dollars.

⇒ Real GDP measured in National income accounts at prices of 2005.

all goods grew proportionately. However, when the price of one good grows faster than the price of another, consumers shift purchases away from the now relatively more expensive good toward the less expensive one. The use of *chain-weighted* indexes helps correct for changes in the market basket.[11]

NOMINAL GDP:

Measures value of output in a given period in the prices of that period or current dollars.

INFLATION

Inflation → rate of change in prices

⇒ Price level is cumulation of past inflations.

P_{t-1} ⇒ last year price level

P_t ⇒ today price level

$$\boxed{\pi = \frac{P_t - P_{t-1}}{P_{t-1}}}$$

inflation

$$\boxed{P_t = P_{t-1} + \pi \times P_{t-1}}$$

⇒ Deflation ⇒ Negative inflation rate

PRICE INDEX

Main price indexes
1. GDP deflator
2. Consumer Price Index
3. Personal consumption expenditure deflator
4. Producer price Index

GDP Deflator

$$\boxed{\text{GDP deflator} = \frac{\text{Nominal GDP of that year}}{\text{Real GDP of that year}}}$$

Consumer Price Index

Measures cost of buying a fixed basket of goods & services representative of purchases of urban consumers.

How is it different ?

The *consumer price index (CPI)* measures the cost of buying a fixed basket of goods and services representative of the purchases of urban consumers. The CPI differs in three main ways from the GDP deflator. First, the deflator measures the prices of a much wider group of goods than the CPI does. Second, the CPI measures the cost of a given basket of goods, which is the same from year to year. The basket of goods included in the GDP deflator, however, differs from year to year, depending on what is produced in the economy in each year. When corn crops are large, corn receives a relatively large weight in the computation of the GDP deflator. By contrast, the CPI measures the cost of a fixed basket of goods that does not vary over time. Third, the CPI directly includes prices of imports, whereas the deflator includes only prices of goods *produced* in the United States.[13]

Personal Consumption Expenditure Deflator

→ Measures inflation in consumer purchases based on consumption sector of national income accounts.

→ Chain weighted index.

→ More important

Producer Price Index

→ Measure of cost of given basket of goods.
→ Differs from CPI in coverage of goods.
→ Includes raw materials and semi finished goods
→ designed to measure prices at early stage of distribution system.

→ CPI measures retail level of buying.

→ flexible

→ serve as business cycle indicators

Core Inflation

→ does not include food & energy prices as they are volatile.
→ reported for PCE and CPI deflator.

Unemployement

→ UNEMPLOYEMENT RATE:
fraction of workforce that is out of work & looking for a job or expecting recall from layoff.

→ 4% okayish
9% very high

→ High unemployement ⟺ Periods of Recession

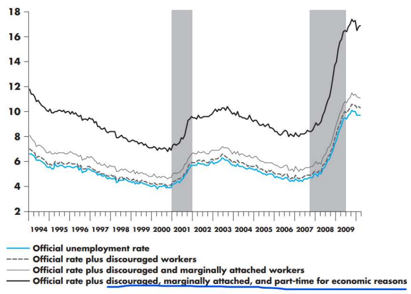


FIGURE 2-9 ALTERNATIVE MEASURES OF UNEMPLOYMENT RATES, 1994–2010.
(Source: Bureau of Labor Statistics.)

Interest Rates

The interest rate states the rate of payment on a loan or other investment, over and above principal repayment, in terms of an annual percentage. If you have $1,000 in the

1. The interest rate

2. These rates differ according to the creditworthiness of the borrower, the length of the loan, and many other aspects of agreement between borrower and lender. (Some of the

3. Treasury Bills among most heavily traded assets

4. Nominal Interest Rates → returns in dollars
Real Interest Rates → subtract inflation to give return in terms of dollars in constant value
safer

EXCHANGE RATES

In the United States, things monetary are measured in U.S. dollars. Canada uses Canadian dollars. Much of Europe uses the euro. The *exchange rate* is the price of foreign currency. For example, the exchange rate with the Japanese yen (April 2010) is a little bit more than one U.S. cent. The British pound is worth about US$1.53. Some countries allow their exchange rates to *float*, meaning the price is determined by supply and demand. Both Japan and Britain follow this policy, so their exchange rates fluctuate over time. Other countries *fix* the value of their exchange rate by offering to exchange their currency for dollars at a fixed rate. For example, the Bermuda dollar is always worth exactly one U.S. dollar and the Hong Kong dollar is set at US$0.13. In practice, many countries intervene to control their exchange rates at some times but not at others, so their exchange rates are neither purely fixed nor purely floating.

Ch- 9

Aggregate Demand

$$\boxed{\text{Aggregate demand} = C + I + G + NX}$$

Equilibrium level ⇒ Output = Aggregate Demand

$= Y = AD = C + I + G + NX$

When $AD \neq Y$

⇒ Unplanned Inventory investment or disinvestment.

$$\boxed{IU = Y - AD}$$

$Y > AD \Rightarrow IU > 0$

As excess inventory accumulates, firm cut back on production until output & aggregate demand are again in equilibrium.

$Y < AD \Rightarrow IU < 0$

Inventories are drawn down until demand is met.

Consumption function

Using $G = 0$
$NX = 0$

① Consumption of goods ∝ Income

Consumption function:
Relationship b/w consumption & income

$$\boxed{C = \bar{C} + cY}$$

intercept
level of consumption when income = 0
$\bar{C} > 0$

$0 < c < 1$
slope
for every 1\$ increase in income, consumption increases by c\$
called $\boxed{\text{Marginal Propensity to consume (MPC)}}$

Consumption And Saving

$S = Y - C$ → consumption

income

$= Y - (\bar{C} + cY)$

$$\boxed{S = Y(1-c) - \bar{C}}$$

Margin Propensity to save.

$1 - c > 0$
⇒ increasing function

$$\boxed{S = Y(1-c) - \bar{C}}$$ $\Rightarrow$ increasing function

margin Propensity to save.

With Government & Investment.

Government, taxes $\rightarrow$ autonomous $\rightarrow$ not something we control

Investment $\rightarrow \bar{I}$
Govt Spending $\rightarrow \bar{G}$
Taxes $\rightarrow \overline{TA}$
Transfers $\rightarrow \overline{TR}$
Net Exports $\rightarrow \overline{NX}$

Disposable Income $= Y - TA + TR$

$\Rightarrow$ consumption now depends on Disposable Income.

$$\boxed{\bar{C} + c(YD) = \bar{C} + c(Y - TA + TR)}$$

$$\text{Aggregate Demand} = C + \bar{I} + \bar{G} + \overline{NX}$$
$$= \bar{C} + c(Y - \overline{TA} + \overline{TR}) + \bar{I} + \bar{G} + \overline{NX}$$
$$= \left[\bar{C} - c(\overline{TA} - \overline{TR}) + \bar{I} + \bar{G} + \overline{NX}\right] + cY$$

$$\boxed{\text{Aggregate Demand} = \bar{A} + cY}$$ agg. demand inc with level of income

$$\boxed{\bar{A} = \bar{C} - c(\overline{TA} - \overline{TR}) + \bar{I} + \bar{G} + \overline{NX}}$$

independent of level of income.

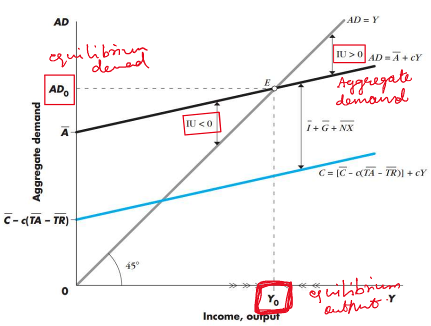


FIGURE 9-2 DETERMINATION OF EQUILIBRIUM INCOME AND OUTPUT.

Equilibrium Point when Agg. demand = Output.

$$x = \bar{A} + cx$$
$$x(1-c) = \bar{A}$$

$$\boxed{\text{equilibrium output } Y_0 = \frac{\bar{A}}{1-c}}$$ → autonomous spending

$$\boxed{\begin{array}{l} Y_0 \propto \bar{A} \\ Y_0 \propto \frac{1}{c} \end{array}}$$

$$\boxed{\Delta Y_0 = \frac{1}{1-c}\,\Delta \bar{A}}$$

Savings And Investment.

During equilibrium, Agg. demand = Output.

$$\boxed{\text{Planned investment} = \text{Saving}}$$ ($G=0$, $NX=0$).

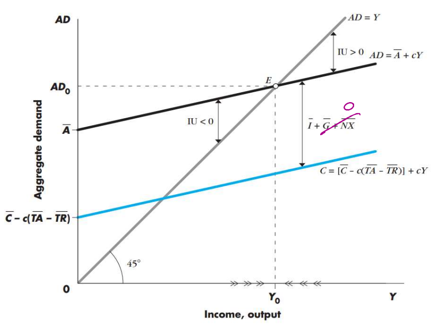


FIGURE 9-2 DETERMINATION OF EQUILIBRIUM INCOME AND OUTPUT.

consump.

$$C = \bar{C} + c(Y + TR - TA)$$

$$S = Y - C = Y - (\bar{C} + c(Y + TR - TA))$$

$$= \cancel{Y} - (\bar{C} + \cancel{cY} + cTR - cTA)$$

$$\boxed{S = Y(1-c) + c(TA-TR) - \bar{C}}$$

$$Y_0 = \frac{\bar{A}}{1-c} = \frac{\bar{C} - c(TA-TR) + \bar{I} + \cancel{\bar{G}} + NX}{1-c}$$

$$= \cancel{\bar{C}} - \cancel{c(TA-TR)} + \bar{I} + \cancel{c(TA-TR)} - \cancel{\bar{C}}$$

$$\boxed{S = \bar{I}}$$

The equality between saving and investment can be seen directly from national income accounting. Since income is either spent or saved, $Y = C + S$. Without government and foreign trade, aggregate demand equals consumption plus investment, $Y = C + I$. Putting the two together, we have $C + S = C + I$, or $S = I$.

If we include government and foreign trade in the analysis, we get a more complete picture relating investment to saving and also to net exports. Now income can either be spent, saved, or paid in taxes, so $Y = C + S + TA - TR$ and complete aggregate demand is $Y = C + I + G + NX$. Therefore,

$$\begin{aligned} C + I + G + NX &= C + S + TA - TR \\ I &= S + (TA - TR - G) - NX \end{aligned} \tag{14}$$

That is, investment equals private savings (S) plus the government budget surplus ($TA - TR - G$) minus net exports (NX), or plus net imports if you prefer.

That is, investment equals private savings (S) plus the government budget surplus ($TA - TR - G$) minus net exports (NX), or plus net imports if you prefer.

How much does 1\$ increase in autonomous spending increase in income?

1\$ increase in autonomous spending → 1\$ increase in income → consumption increases $C = \bar{c} + cY$ → $c\Delta Y$ → 1\$ increase in autonomous spending

$\bar{A}$	Increase in production	Increase In income
$\Delta\bar{A}$	$\Delta\bar{A}$	$\Delta\bar{A}$
$c\Delta\bar{A}$	$c\Delta\bar{A}$	$(1+c)\Delta\bar{A}$
$c^2\Delta A$	$c^2\Delta A$	$(1+c+c^2)\Delta A$ ...

$$(1+c+c^2+\dots)\Delta A$$

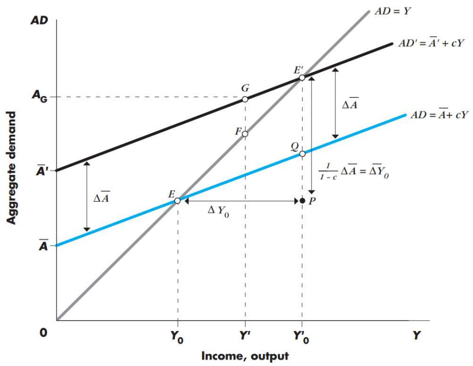
$$= \boxed{\boxed{\frac{\Delta\bar{A}}{1-c}} = \Delta Y_0 = \Delta AD_0}$$

$$\boxed{\frac{1}{1-c} = \text{multiplier}}$$

From equation (16), therefore, we find that the cumulative change in aggregate spending is equal to a multiple of the increase in autonomous spending—just as we deduced from equation (12). The multiple $1/(1-c)$ is called the *multiplier*.[3] **The multiplier is the amount by which equilibrium output changes when autonomous aggregate demand increases by 1 unit.**

Output changes with change in $\bar{A}$ BUT change in output can be larger than change in $\bar{A}$.

The magnitude of the income change required to restore equilibrium depends on two factors. The larger the increase in autonomous spending, represented in Figure 9-3 by the parallel shift in the aggregate demand schedule, the larger the income change. Furthermore, the larger the marginal propensity to consume—that is, the steeper the aggregate demand schedule—the larger the income change.



GOVT SECTOR

fiscal Policy:
Policy of government with regard to level of govt purchases, level of transfers & tax structure.

Assume $\underbrace{G = \bar{G} \quad , \quad TR = \overline{TR}}_{\text{costant}}$, $TA = tY$
$\downarrow$ fraction of income

Rewriting consumption function

$$C = \bar{C} + c(YD)$$
$$= \bar{C} + c(Y + TR - TA)$$
$$= \bar{C} + c(Y - tY + TR)$$
$$C = \bar{C} + c(Y(1-t) + TR)$$

$$\boxed{C = \bar{C} + c\overline{TR} + cY(1-t)}$$

New $\boxed{c = c(1-t)}$
$\downarrow$ new marginal propensity to consume.

$$AD = C + I + G + NX$$
$$= \bar{C} + c\overline{TR} + cY(1-t) + \boxed{\bar{I} + \bar{G} + \overline{NX}}$$
$$= (\bar{C} + c\overline{TR} + \bar{I} + \bar{G} + \overline{NX}) + cY(1-t)$$

$$\boxed{AD = \bar{A} + c(1-t)Y}$$

$$\bar{A} = \bar{C} + c\overline{TR} + \bar{I} + \bar{G} + \overline{NX}$$

EQUILIBRIUM INCOME

$$Y = AD$$

$$x(1-c(1-t)) = \bar{A}$$

equilibrium income $\boxed{Y_0 = \dfrac{\bar{A}}{1-c(1-t)}}$

lowers the multiplier

Income Taxes as Automatic Stabilizers

The proportional income tax is one example of the important concept of *automatic stabilizers*. **As you remember, an automatic stabilizer is any mechanism in the economy that automatically—that is, without case-by-case government intervention—reduces the amount by which output changes in response to a change in autonomous demand.**

⇒ Investment demand have a smaller effect on output when automatic stabilizers such as income tax which reduces the multiplier are in place.

Unemployment benefits → enable unemployed to consume even when they don't have a job

→ TR↑, Y↓

→ Demand falls less when someone becomes unemployed and receives benefits than it would if there are NO Benefits.

→ This makes multiplier smaller & output more stable

Effects of Change on fiscal Policy.

1. Increase in Govt spending

CHAPTER 9•INCOME AND SPENDING

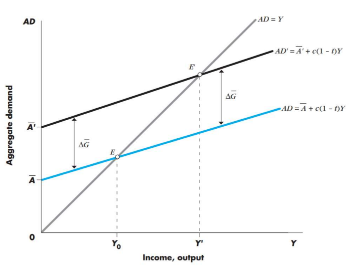


FIGURE 9-4 THE EFFECTS OF AN INCREASE IN GOVERNMENT PURCHASES.

$$\boxed{\Delta Y_0 = \Delta \bar{G} + c(1-t)\Delta Y_0}$$

$$\Delta Y_0 = \frac{\Delta \bar{G}}{1-c(1-t)}$$

$$\boxed{\alpha_G = \frac{1}{1-c(1-t)}}$$

$$\boxed{\Delta Y_0 = \alpha_G \Delta \bar{G}}$$

Suppose 1$ increase in govt purchase

$c = 0.8$

$t = 0.25$

$$\Delta Y_0 = \frac{1\,(1)}{1-0.8(0.75)}$$

$c = 0.8$

$t = 0.25$

$$\Delta Y_0 = \frac{1\,(1)}{1 - 0.8(0.75)}$$

$$= \frac{1}{1 - \frac{8}{10} \times \frac{3}{4}}$$

$$= \frac{1}{1 - \frac{24}{40}} = \frac{40}{16} = \frac{10}{4} = 2.5$$

(2) Increase in $\overline{TR}$

$$\Delta \overline{A} = c\Delta \overline{TR}$$

$$\boxed{\Delta Y_0 = \frac{c\Delta TR}{1 - c(1-t)}}$$

(3) Increase in $\overline{TA}$

If the government raises marginal tax rates, two things happen. The direct effect is that aggregate demand will be reduced since the increased taxes reduce disposable income and therefore consumption. In addition, the multiplier will be smaller, so shocks will have a smaller effect on aggregate demand.

Since the theory we are developing implies that changes in government spending and taxes affect the level of income, it seems that fiscal policy can be used to stabilize the economy. When the economy is in a recession, or growing slowly, perhaps taxes should be cut or spending increased to get output to rise. And when the economy is booming, perhaps taxes should be increased or government spending cut to get back down to full employment. Indeed, fiscal policy is used actively to try to stabilize the economy, as in 2009, when the Obama administration cut taxes and massively increased spending in order to fight the Great Recession.

BUDGET

$$\boxed{\text{Budget Surplus} = TA - \overline{G} - \overline{TR}}$$

excess of govt revenues, taxes over its total expenditures consisting of purchases of goods & services & transfer payments.

Negative Budget Surplus/ Budget Deficit $\Rightarrow$ Expenditure > Revenue collected

$$BS = tY - \overline{G} - \overline{TR}$$

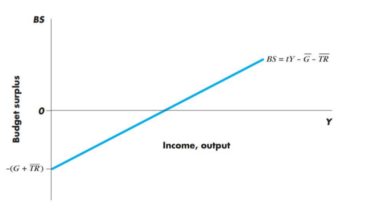


FIGURE 9-6 THE BUDGET SURPLUS.

Budget depends — t, $\overline{G}$, $\overline{TR}$

FIGURE 9-6 THE BUDGET SURPLUS.

Budget depends — t, $\overline{G}$, $\overline{TR}$, anything that shifts income

Suppose increase in investment ⇒ increase in output ⇒ Budget deficit will fall / surplus ↑ because tax revenues has increased.

Effects of Govt Purchases & Tax Changes on BS

Increased Govt Purchases $\quad \Delta Y_0 = \alpha_G \Delta \overline{G}$

Increase in tax revenue $\quad \Delta TA = t\alpha_G \Delta \overline{G}$

$$\Delta BS = \Delta TA - \Delta \overline{G}$$

$$\Delta BS = t\alpha_G \Delta \overline{G} - \Delta \overline{G}$$

$$\Delta BS = \Delta G(t\alpha_G - 1)$$

$$= \Delta G\left(t\frac{1}{1-(1-t)c} - 1\right)$$

$$\Delta BS = \left(\frac{t - (1-(1-t)c)}{1-(1-t)c}\right)\Delta G$$

$$= \left(\frac{t - 1 + c(1-t)}{1-(1-t)c}\right)\Delta G$$

$$\boxed{\Delta BS = \frac{\underbrace{(c-1)}_{-ve}\overbrace{(1-t)}^{+ve}}{1-(1-t)c}\Delta G}$$

$$\boxed{\Delta BS < 0}$$

Increase of tax rate (t)

$$YD = Y - TA + TR$$

$$YD = Y(1-t) + TR \qquad \boxed{\Delta Y_0 = -Y\Delta t}$$

$$\Delta BS = \Delta TA - \cancel{\Delta G}^{\,0}$$

$$\Delta BS = t\Delta Y_0 + \Delta t Y_0$$

$$= t(-Y\Delta t) + \Delta t\, Y_0$$

$$\boxed{\Delta BS = Y_0 \Delta t(1-t)}$$

Balanced Budget Multiplier

$$\Delta TA = \Delta \bar{G}$$

$$t\Delta Y + Y\underbrace{\Delta t}_{x} = \underbrace{\Delta \bar{G}}_{x}$$

$$t\Delta Y + Yx = x$$

$$\boxed{\Delta Y = \frac{x(1-Y)}{t}}$$

$$\boxed{\Delta Y = \frac{\Delta G(1-Y)}{t}}$$

???

We mention here another interesting result known as the *balanced budget multiplier*. Suppose government spending and taxes are raised in equal amounts and thus in the new equilibrium the budget surplus is unchanged. By how much will output rise? The answer is that for this special experiment the multiplier is equal to 1—output rises by the increase in government spending and no more.

Full Employement Budget Surplus

⇒ Increase in budget surplus may not be due to changes in fiscal policy

BS^* full employement budget surplus ⇒ Measures budget surplus at full employement level of income or at potential output.

$Y^* \to$ full employement level of income

$$\boxed{BS^* = tY^* - \bar{G} - \overline{TR}}$$

other names:

1. high employment surplus
2. standardized surplus
3. cyclically adjusted surplus
4. structural surplus

$$\boxed{BS^* - BS = t(Y^* - Y)}$$

1. actual output < full emp. $BS^* > BS$.
2. actual output > full empl. output $BS > BS^*$

plus is less than the actual surplus. The difference between the actual and the full-employment budget is the *cyclical* component of the budget. In a recession the cyclical component tends to show a deficit, and in a boom there may even be a surplus.

Excercises

Q1 $C = 100 + 0.8Y$

Exercises

01

$C = 100 + 0.8Y$

$I = 50$

(a) $Y = 100 + 0.8Y + 50$

$\frac{\cancel{2}}{10} Y = \cancel{150}^{\,75}$

$\boxed{Y = 750}$

(b) $S = I = \boxed{50}$

(c)

$AD = 100 + 0.8 \times 800 + 50$

$= 100 + 640 + 50$

$= \boxed{790}$

$\boxed{IU = 10}$

(d) $I = 100$

$Y = 100 + 0.8Y + 100$

$0.2Y = 200$

$\boxed{Y = 1000}$

(e) $\alpha = \frac{1}{1-c} = \frac{1}{0.2} = 5$

CH-10

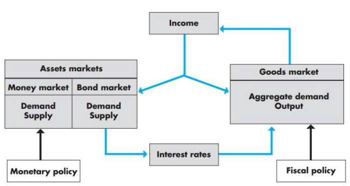


FIGURE 10-3 THE STRUCTURE OF THE *IS-LM* MODEL.

The IS-LM model emphasizes the interaction between the goods and assets markets. Spending, interest rates, and income are determined jointly by equilibrium in the goods and assets markets.

IS CURVE

Shows combinations of interest rates & level of output such that planned spending equals income.

Investment demand Schedule

Higher interest rates for borrowing → less profit → less willing to invest

Higher interest rates for borrowing money for investing → less profit → less willing to invest
$I \downarrow$

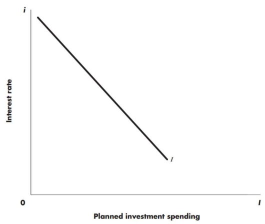
autonomous investment spending → independent of
Income
Rate of interest

Investment spending function

$$\boxed{I = \bar{I} - bi} \qquad \underline{b > 0}$$

responsiveness of investment

interest rate

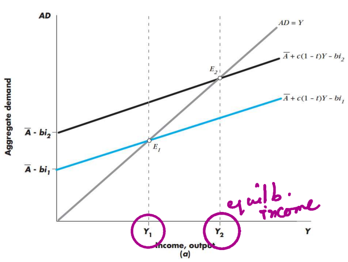


$$AD = C + I + G + NX$$

$$= \left[\bar{C} + c\overline{TR} + c(1-t)\bar{Y}\right] + \left(\bar{I} - bi\right) + \bar{G} + \overline{NX}$$

$$\boxed{AD = \bar{A} + c(1-t)\bar{Y} - bi}$$

$$\boxed{\bar{A} = \bar{C} + c\overline{TR} + \bar{I} + \bar{G} + \overline{NX}}$$



IS curve → also called goods market equilibrium schedule.

$$Y_0 = \overline{AD} = \bar{A} + (1-t)cY_0 - bi$$

$$Y_0\left(1 - (1-t)c\right) = \bar{A} - bi$$

$$\boxed{Y_0 = \frac{\bar{A} - bi}{1 - (1-t)c}}$$

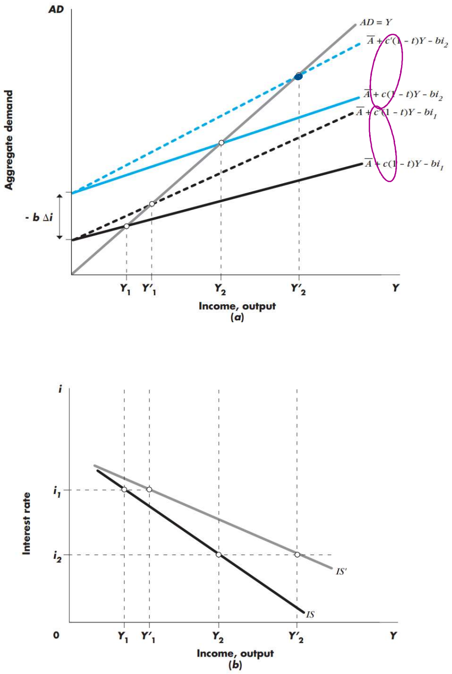


FIGURE 10-6 EFFECT OF THE MULTIPLIER ON THE SLOPE OF THE IS CURVE.
A higher marginal propensity to spend results in a steeper aggregate demand curve and, consequently, a flatter IS curve.

aggregate demand at different multipliers.

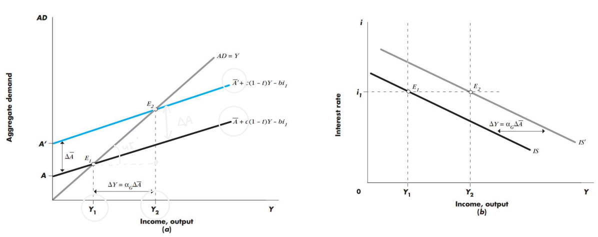
$b \downarrow, \frac{1}{1-(1-t)c} \downarrow \Rightarrow$ slope $\uparrow$

$$\bar{Y} = (\bar{A} - bi)\alpha_G$$

$$i = \frac{A - \frac{\bar{Y}}{\alpha_G}}{b} \Rightarrow \boxed{i = \frac{\bar{A}}{b} - \frac{Y}{\alpha_G b}}$$

$\alpha_G \rightarrow$ tax rate

$\uparrow t \longrightarrow \uparrow$ slope.



LM CURVE

Shows combinations of interest rates & levels of output such that money demand equals money supply.

① Demand for money ⇒ demand for real money balances

Real vs Nominal money.

At this stage we have to reinforce the crucial distinction between *real and nominal* variables. The nominal demand for money is the individual's demand for a given number of dollars. Similarly, the nominal demand for bonds is the demand for a given number of dollars' worth of bonds. The real demand for money is the demand for money expressed in terms of the number of units of goods that money will buy: It is equal to the nominal demand for money divided by the price level. If the nominal demand for money is \$100 and the price level is \$2 per good—meaning that the representative basket of goods cost \$2—the real demand for money is 50 goods. If the price level later doubles to \$4 per good and the demand for nominal money likewise doubles to \$200, the real demand for money is unchanged at 50 goods.

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Real money balances—real balances, for short—are the quantity of nominal money divided by the price level. The real demand for money is called the *demand for real balances.*

(2) Demand for real balances → level of real income → individuals pay for their purchases which in turn depend on income

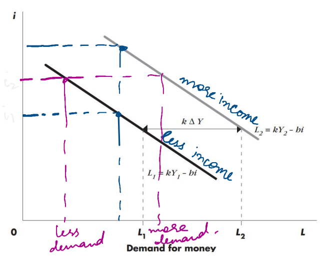
→ interest rate → cost of holding money (interest that is forgone by holding money rather than other assets).

$\boxed{L = kY - hi}$ $k, h > 0$

k: sensitivity of demand for real balances to level of income.

h: same thing for interest rates

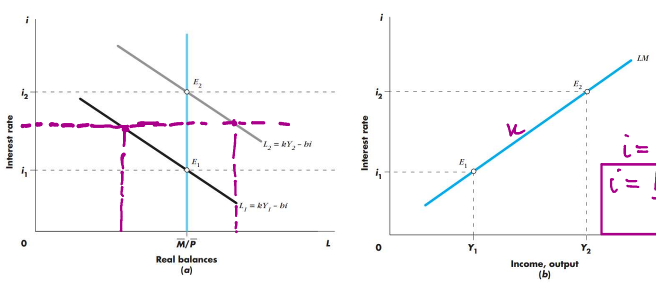
↑ interest rate ⇒ more costly to hold money ⇒ less cash will be held at each level of income



Nominal Quantity of money → M
Nominal quantity of money as given at the level → $\bar{M}$

Price level (constant) → $\bar{P}$

$\boxed{\text{Real Money supply} \to \frac{\bar{M}}{\bar{P}}}$



$i = kY - L$

$\boxed{i = \frac{kY}{b} - \frac{L}{b}}$

The *LM* schedule, or *money market equilibrium schedule*, shows all combinations of interest rates and levels of income such that the demand for real balances is equal to the supply. Along the *LM* schedule, the money market is in equilibrium.

$$\frac{\left(\bar{A} - \frac{Y}{\alpha_G}\right)}{b} = \frac{kY - \frac{\bar{M}}{\bar{P}}}{h}$$

(a)

Income, output (b)

The *LM* schedule, or *money market equilibrium schedule*, shows all combinations of interest rates and levels of income such that the demand for real balances is equal to the supply. Along the *LM* schedule, the money market is in equilibrium.

Increase in interest rate ⇒ Has to meet fixed demand for real balances ⇒ Income HAS To increase

$Y = (\bar{A} - bi)\alpha_G$

$\frac{\bar{M}}{\bar{P}} = kY - hi$

fixed

$\boxed{i = \frac{1}{h}\left(kY - \frac{\bar{M}}{\bar{P}}\right)}$

$\frac{\bar{M}}{\bar{P}} = kY - hi$

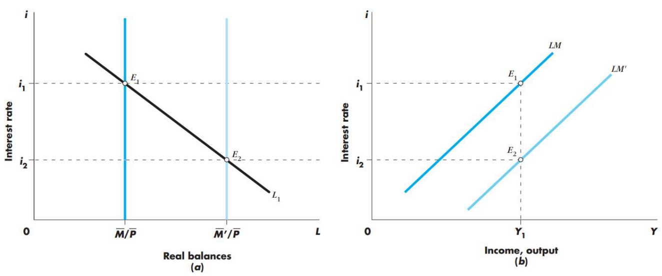


FIGURE 10-10 AN INCREASE IN THE SUPPLY OF MONEY SHIFTS THE *LM* CURVE TO THE RIGHT.

$\frac{\bar{A} - bi}{1-(1-t)c} = \frac{\frac{\bar{M}}{\bar{P}} + hi}{k}$

$\frac{k(\bar{A} - bi)}{1-(1-t)c} = \frac{\bar{M}}{\bar{P}} + hi$

$k(\bar{A} - bi) = \left(\frac{\bar{M}}{\bar{P}} + hi\right)(1-(1-t)c)$

$k\bar{A} - kbi = \frac{\bar{M}}{\bar{P}} - \frac{\bar{M}}{\bar{P}}(1-t)c + hi(1-(1-t)c)$

$k\bar{A} - \frac{\bar{M}}{\bar{P}} + \frac{\bar{M}}{\bar{P}}(1-t)c = i\left(h - hc(1-t) + kb\right)$

$\boxed{\frac{k\bar{A} - \frac{\bar{M}}{\bar{P}} + \frac{\bar{M}}{\bar{P}}(1-t)c}{h - hc(1-t) + kb} = i}$

LM-IS curve

$\frac{k\bar{A} - \frac{\bar{M}}{\bar{P}} + \frac{\bar{M}}{\bar{P}}\left(\frac{\alpha-1}{\alpha}\right)}{h - h\left(\frac{\alpha-1}{\alpha}\right) + kb}$

$\frac{1}{1-x} = \alpha_G$

$\frac{1}{\alpha_G} = 1-x$

$1 - \frac{1}{\alpha_G} = x$

$\frac{\alpha_G - 1}{\alpha_G}$

$\frac{\alpha k\bar{A} - \alpha\frac{\bar{M}}{\bar{P}} + \frac{\bar{M}}{\bar{P}}(\alpha-1)}{\alpha h - h(\alpha-1) + kb\alpha}$

$\frac{\alpha k\bar{A} - \frac{\bar{M}}{\bar{P}}}{h + k\alpha b}$

$\boxed{i = \frac{k}{h}\gamma\bar{A} - \frac{\gamma}{\alpha h}\frac{\bar{M}}{\bar{P}}}$

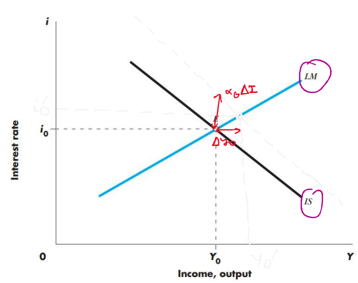


FIGURE 10-11 GOODS AND MONEY MARKET EQUILIBRIUM.

At point E, interest rates and income levels are such that the public holds the existing money stock and planned spending equals output.

It is worth stepping back now to review our assumptions and the meaning of the equilibrium at *E*. The major assumption is that the price level is constant and that firms are willing to supply whatever amount of output is demanded at that price level. Thus, we assume that the level of output Y_0 in Figure 10-11 will be willingly supplied by firms at the price level $\bar{P}$. We repeat that this assumption is one that is temporarily needed for the development of the analysis; it corresponds to the assumption of a flat, short-run aggregate supply curve.

$\boxed{h\bar{A} - \frac{hY}{\alpha_G} = bkY - \frac{b\bar{M}}{\bar{P}}}$

$h\bar{A} + \frac{b\bar{M}}{\bar{P}} = Y\left(bk + \frac{h}{\alpha_G}\right)$

$\boxed{\frac{h\bar{A} + \frac{b\bar{M}}{\bar{P}}}{bk + \frac{h}{\alpha_G}} = Y}$

$\frac{\left(h\bar{A} + \frac{b\bar{M}}{\bar{P}}\right)\alpha_G}{bk\alpha_G + h} = Y$

$\frac{h\alpha_G\bar{A} + b\alpha_G\frac{\bar{M}}{\bar{P}}}{bk\alpha_G + h} = Y$

$\Rightarrow \boxed{\gamma\bar{A} + \gamma\frac{b}{h}\frac{\bar{M}}{\bar{P}} = Y}$

$\boxed{\gamma = \frac{\alpha_G h}{bk\alpha_G + h}}$

$\boxed{\gamma = \frac{\alpha_G h}{bk\alpha_G + h} = \text{fiscal policy multiplier}}$

Change in IS curve by increased autonomous spending

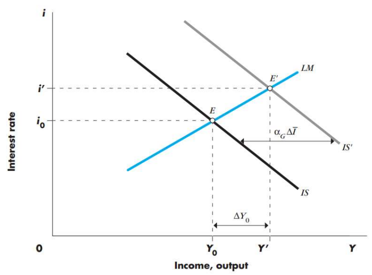
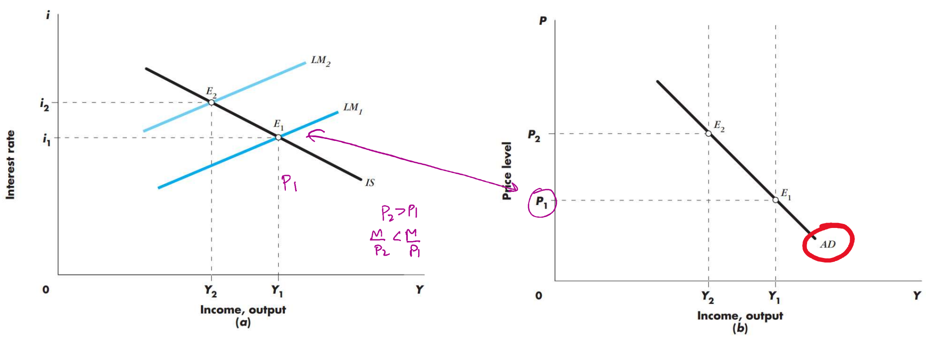


FIGURE 10-12 AN INCREASE IN AUTONOMOUS SPENDING SHIFTS THE *IS* CURVE TO THE RIGHT.
The equilibrium interest rate and level of income both rise.

But what is the economics of what is happening? The increase in autonomous spending does tend to increase the level of income. But an increase in income increases the demand for money. With the supply of money fixed, the interest rate has to rise to ensure that the demand for money stays equal to the fixed supply. When the interest rate rises, investment spending is reduced because investment is negatively related to the interest rate. Accordingly, the equilibrium change in income is less than the horizontal shift of the *IS* curve, $\alpha_G \Delta\bar{I}$.

AGGREGATE DEMAND SCHEDULE



$$\frac{\Delta Y}{\Delta G} = \gamma \qquad \gamma = \frac{h\alpha_G}{h + kb\alpha_G}$$

$$i = \gamma \frac{k}{h}\bar{A} - \gamma \frac{1}{h\alpha_G}\frac{\overline{M}}{P}$$

$$Y = \frac{h\alpha_G}{h + kb\alpha_G}\bar{A} + \frac{b\alpha_G}{h + kb\alpha_G}\frac{\overline{M}}{P}$$

The *fiscal policy multiplier* shows how much an increase in government spending changes the equilibrium level of income, holding the real money supply constant. Examine equation (8) and consider the effect of an increase in government spending on income. The increase in government spending, $\Delta\overline{G}$, is a change in autonomous spending, so $\Delta\overline{A} = \Delta\overline{G}$. The effect of the change in $\overline{G}$ is given by

$$\frac{\Delta Y}{\Delta \overline{G}} = \gamma \qquad \gamma = \frac{h\alpha_G}{h + kb\alpha_G} \tag{10}$$

The expression γ is the fiscal or government spending multiplier once interest rate adjustment is taken into account. Consider how this multiplier, γ, differs from the simpler expression α_G that applied under constant interest rates. Inspection shows that γ is less than α_G since $1/(1 + k\alpha_G b/h)$ is less than 1. This represents the dampening effect of increased interest rates associated with a fiscal expansion in the *IS-LM* model.

We note that the expression in equation (10) is almost zero if h is very small and that it is equal to α_G if h approaches infinity. This corresponds, respectively, to vertical and horizontal *LM* schedules. Similarly, a large value of either b or k serves to reduce the effect of government spending on income. Why? A high value of k implies a large increase in money demand as income rises and hence a large increase in interest rates required to maintain money market equilibrium. In combination with a high b, this implies a large reduction in private aggregate demand.

THE MONETARY POLICY MULTIPLIER

The *monetary policy multiplier* shows how much an increase in the real money supply increases the equilibrium level of income, keeping fiscal policy unchanged. Using equation (8) to examine the effects of an increase in the real money supply on income, we have

$$\frac{\Delta Y}{\Delta(\overline{M}/\overline{P})} = \frac{b}{h}\gamma = \frac{b\alpha_G}{h + kb\alpha_G} \tag{11}$$

The smaller h and k and the larger b and α_G, the more expansionary the effect of an increase in real balances on the equilibrium level of income. Large b and α_G correspond to a very flat *IS* schedule.